

Autonomous Blood Suction for Minimally Invasive Surgeries

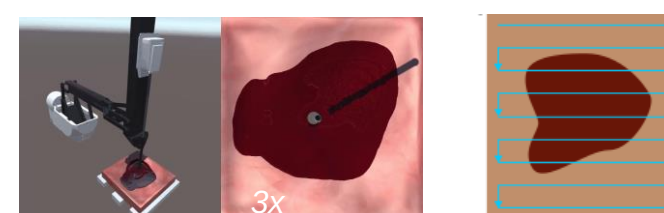
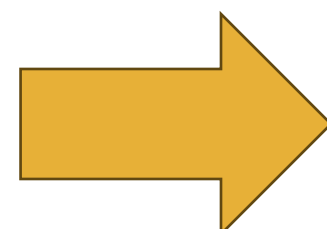
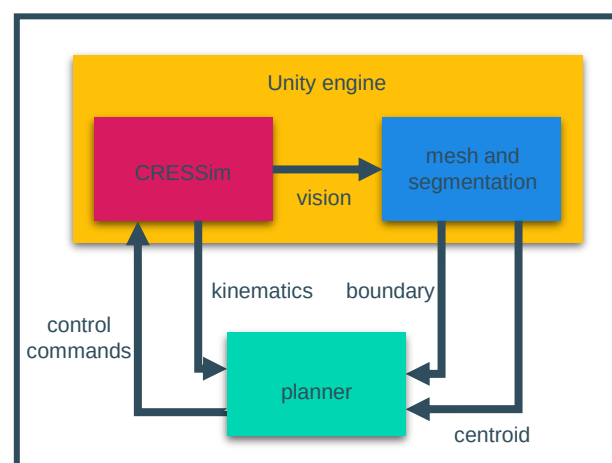
Matthew Foster¹, Charles DeLorey², and Dominic Jones²

¹School of Mechanical Engineering, University of Leeds, Leeds, UK

²STORM Lab, School of Electronic and Electrical Engineering, University of Leeds, Leeds, UK

Clinical Motivation:

- Bleeds during surgery **interrupt surgery progress** and **obscure surgeon view**, necessitate suction
- Prior work has explored haemostasis and blood detection, **optimal trajectory evaluation remains underexplored**
- Automation of blood suction** would allow surgeons to focus on procedure, lighten cognitive load



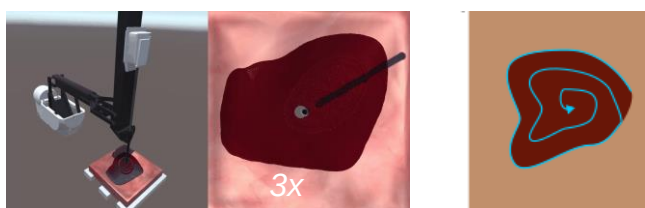
Global Zigzag



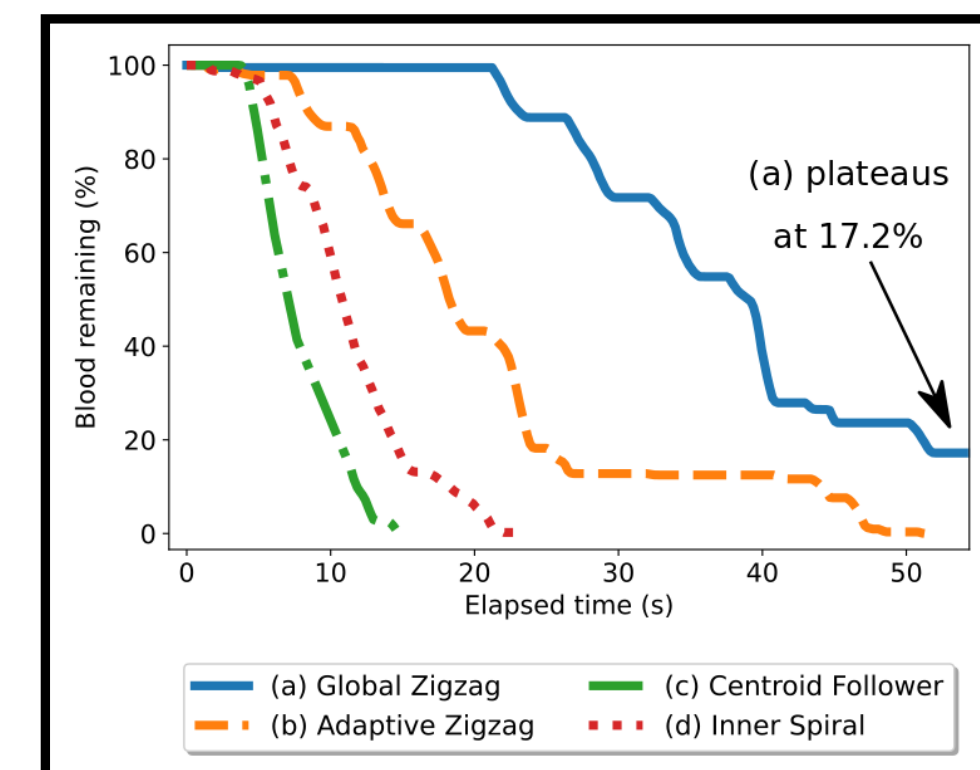
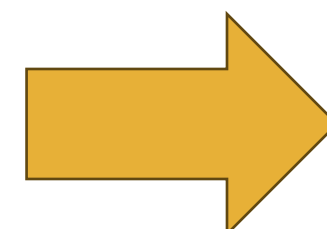
Adaptive Zigzag



Centroid Follower



Inward Spiral



Methods:

- Simulation in CRESSim [1] with dVRK [2] (XY plane only)
- Blood detection used HSV thresholding with OpenCV to generate binary mask, highlighting the largest blood pool
- OpenCV extraction of blood pool size, centroid, perimeter
- Planner script generated robot path (with prior kinematics) and sent control commands to the simulated robot

Conclusions:

- Three methods achieved <20% blood after 30 s
- Centroid Follower cleared fastest, however would stall on multiple blood pools
- Inward Spiral achieved best balance of speed, sustained performance, and extendibility**
- Future Work would include:
 - Enable Z-axis positioning, closed-loop control
 - Transfer results to real world
 - More anatomically representative